Chapter 7

# Properties and behavior of light

Every morning we see sunlight which makes our day bright. Can you imagine the world without the Sun? Have you ever thought why does the sun glow? Why do the stars shine at night? Why can't we see without the presence of light? It is because light enables us to see material objects.

Figure 7.1: The sun giving light to various places

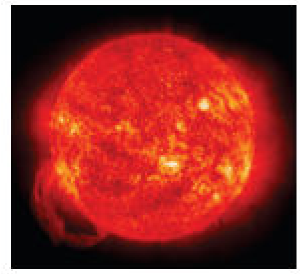
**In this chapter, you will learn about:**

- Luminous and non luminous objects.
- Transparent, opaque and translucent objects.
- Light travels in straight lines.
- Shadow formation.
- Eclipse formation.
- Pinhole camera.
- Phases of moon.

**All the students will be able to:**

- ✓ Understand that light is a form of energy.
- ✓ Know that light travels in a straight line.
- ✓ Differentiate between luminous and non-luminous objects.
- ✓ Identify and differentiate between opaque, translucent and transparent objects in the surroundings.
- ✓ Understand the formation of shadows.
- ✓ Explain how do eclipses occur.
- ✓ Explain the effect of position of an predict location, size and shape of the shadow from the light sources.
- ✓ Explain the working principle of a pinhole camera.
- ✓ Identify different phases of the Moon.

# Luminous and non-luminous objects



Differentiate between luminous and non-luminous objects

Look at the figures on your right.

Which one of these objects shine? Does the sun shine? Does the moon shine?

Your answer may be both. What is the difference between the glowing Sun and the moon light that you see at night?

The Sun shines because it is capable of producing its own light. The Sun is a luminous object. The Moon shines due to the light of the sun. It reflects sun's light. It cannot produce its own light. The Moon is a non-luminous object.

A **luminous** object is one which produces and emits its own light. In other words, it glows on its own accord. To be able to glow, an object must have its own source of energy. The Sun is a luminous object, it is made up of such substances which provide it energy to glow.

A torch is also a luminous object. It shines because of the energy stored in its batteries. Thus we can say that light is a form of energy.

Other examples of luminous objects are lamps, candle, fire flies, stars and lantern fish.

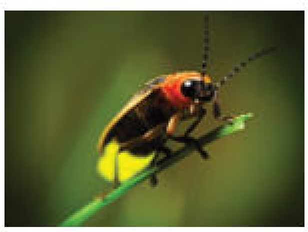
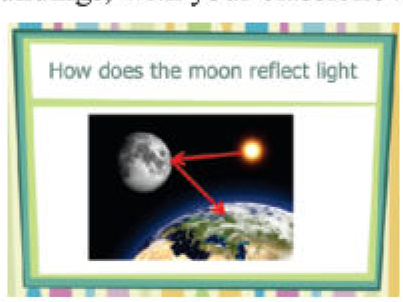


Figure 7.2: Luminous objects

**Do you know?** Our Sun is the principle source of energy for all the organisms on the earth. Our Sun is an atomic furnace that turns mass into energy. Every second, it converts over 657 million tons of hydrogen into 653 million tons of helium. The missing 4 million tons of mass are discharged into space as energy. The Earth receives only about two-billionths of this as heat and light.

A **non-luminous** object is one which does not produce its own light but it reflects light that comes from the luminous object. Some examples of non-luminous objects are furniture in the room, books, clothes, trees and planets. Identify, draw and share names and pictures of luminous and non-luminous objects from your surroundings, with your classfellows.

Figure 7.3: Non-luminous objects



**How are we able to see objects?**

- Know that light helps us to see.
- Explain reflection of light.

Light is a form of energy. When light rays falls on any material object, the light rays bounce back or get reflected. This process is called **Reflection**. Reflection is one of the properties of light. Light from the luminous objects travels in a straight line and directly enters into our eyes.

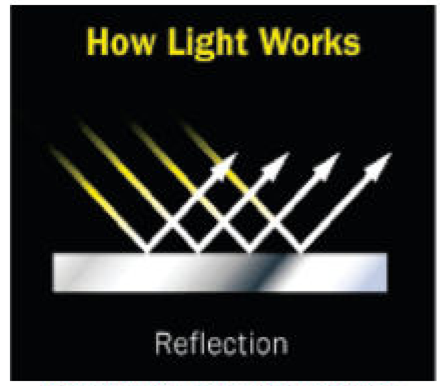
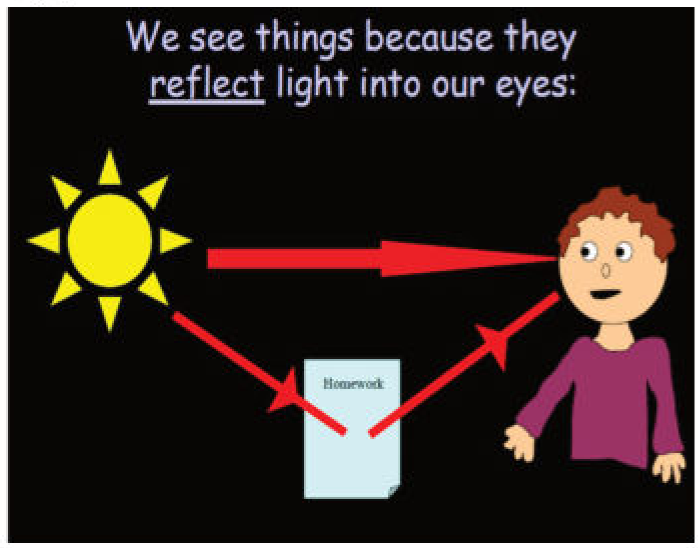


Figure 7.4: Reflection of light

When our eye receives these reflected light rays, we are able to see the objects.

We are not able to see objects in the absence of a light source.




In the dark room, no reflected light enters into our eyes; therefore we cannot see objects in the dark room. Take a torch and a small mirror. Point the torch light towards the mirror. Turn the mirror a little. you get a spot of light on the opposite wall? Observe the bouncing of light by the shining object. Share how we see objects with your classfellows. Do other objects reflect light? Let us study how light behaves with different objects.

## Transparent, translucent and opaque objects

- Identify and differentiate between transparent, opaque and translucent objects in their surroundings.

In playing hide and seek, you often hide yourself under the table or behind any wall. Why don't you hide behind a transparent glass or window? It is obvious that you will be seen if you hide behind a glass because light passes through the glass, as glass is a transparent material.

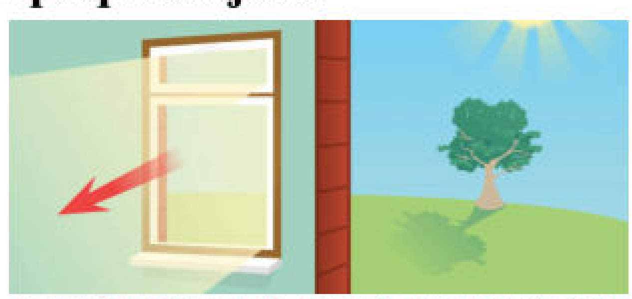


Figure 7.5: Light passing through a transparent window

Light behaves differently with different materials.

**Transparent objects** allow light to travel through them. Materials like air, water, and clear glass are transparent. When the light strikes transparent materials, almost all of it passes directly through them.

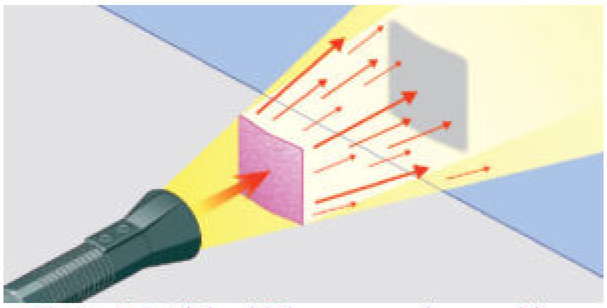


Figure 7.6: Light shining on translucent object

**Translucent objects** allow some light to travel through them. Materials like frosted glass and some plastics are called translucent. When light strikes translucent materials, only some of the light passes through them. The light that passes through these materials does not pass directly through the materials but it is scattered. Therefore, we cannot see clearly through them and objects on the other side of a translucent material appear fuzzy and unclear.

**Opaque objects** do not allow any light to pass through them and block the light. They reflect most of the light and absorb some. Brick wall, wooden objects, tree, thick clothing are opaque materials.

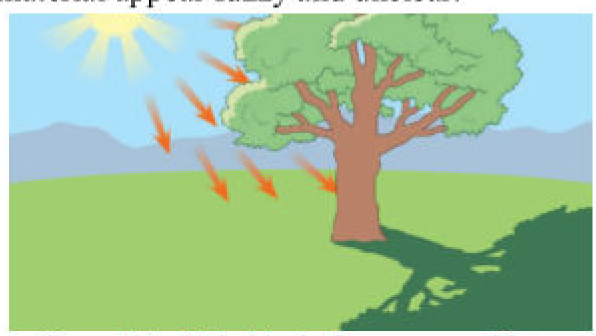


Figure 7.7: Light blocked by opaque objects

**Activity 1:** Investigating transparent, translucent and opaque objects

**What I need:**

- Different objects such as plastic wrap, cloth , paper, butter paper, bulb in which filament is visible, water, reading glasses, sun glasses, woollen clothing, aluminium foil, books, muslin piece, torch.

**What I do:**

1. Take a plastic wrap and shine a torch light on it. Observe whether it allows all light to pass through it, some light to pass through it, or does not allow any light to pass through it. Categorize it in the table according to the behavior of the light.
2. Repeat step 1 with each object.

**What I observed:**

| Transparent objects | Translucent objects | Opaque objects |
|---|---|---|
| | | |

## Activity questions:

1. Which objects are transparent?
2. Which objects are translucent?
3. Which objects are opaque?
4. What makes an object transparent, translucent, or opaque?
5. Ask students to go for an exploration in the school or home for more transparent, translucent and opaque materials. Share the name or the materials with classfellows.

Investigate that light travels in a straight line.

**Light Travels in Straight Lines**

You know that light is a form of energy which helps us see objects. We know that light travels in a straight line. It will continue to travel in a straight line until it strikes an object.

This is one of the properties of light.

**Do you know?**

Light travels very fast . Speed of light in vacuum is 300 million metres per second.

Are you able to see through the bent pipe or a straight pipe?
Obviously through the straight pipe, because light travels in a straight line.

**Activity 2:** Investigating that light travels in a straight line.

**What I need :**

- Three thick cards of 3x5 inches size
- a pencil and a ruler
- a torch
- a cutter or scissors.
- some mounting clips or objects to stand cards.

**What I do:**

1. Draw lines on each card from one corner to the next forming an 'X'.
2. Cut a hole in each card where the lines intersect.
3. Line up the cards with the help of the mounting clips or stand about 15 cm apart in an uneven line on the table, so that holes are not in a straight line.
4. Point light from a luminous object (candle or torch) to pass through and look from the other side. Record your observation in the table below.
5. Now line up the cards in a straight and even line so that holes are in a straight line. Repeat step 4.

**Teacher Note:** Arrange the materials and engage students in performing the activities in groups or demonstrate it.

## What I observed:

| Cards in an even line | Cards in an uneven line |
|---|---|
| Light…. | Light…. |
| Because… | Because… |

## Activity questions:

1. What happens when you allow light to pass through the uneven line of cards?
2. What happens when you allow light to pass through the even line of cards?
3. Is the behavior of light different in both the cases?
4. What would you conclude about the above activity?

## Shadows

Explain the formation of shadows.

Light travels in a straight line until it strikes an object. When light strikes an opaque object, the light is blocked, thus a shadow is formed on the opposite side of the object.

Figure 7.8: Formation of a shadow

If an object is moved further away from the light source, the shadow gets smaller. If an object is moved closer to the light source, the shadow gets bigger.


Figure 7.9: Formation of a big shadow

**Activity 3:** Investigating the effect of the position of an object on the size , shape and location of its shadow.

Investigate the location, size and shape of a shadow from a light source relative to the position of the object.

### What I need:

- a torch
- a dark room
- a doll or any other toy (should be opaque)

### What to do:

1. Keep the table about seven feet away from the wall.
2. Make the room dark.
3. Turn on the torch and place it on the table.
4. Take the toy and bring it closer to the torch. Mark this as position 1.
5. Observe the size, shape and location of the shadow that is formed on the wall and record your observation in the table .
6. Now take the toy a little further away from the torch. Mark this position 2. Observe the shadow and record your observation.
7. Again, take the toy further away from the torch. Mark this as position 3. Observe the shadow and record your observation.

### What I observed:

| Positions of the toy | Shape of the shadow | Size of the shadow | Location of the shadow |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

## Activity questions:

1. At which position is the biggest shadow formed?
2. At which position is the smallest shadow formed?
3. What can you say about the shape and location of the shadows at different positions?
4. What would you conclude about shadow from the above activity?

# Lunar and Solar Eclipses

Explain the formation of eclipses.

A Solar Eclipse occurs when the Moon comes in front of the Sun and blocks most of the Sun's light from reaching the Earth. During a total eclipse, all you can see from the Earth is a ring of light around the Moon, which is a part of the Sun that the Moon does not cover. It is dangerous to look at the solar eclipse directly with un-protected eyes.

A lunar eclipse occurs when the Earth comes in between the Moon and the Sun, thus forming a shadow of the Earth, and the Moon passes through this shadow of the Earth. A lunar eclipse can last up to an hour and a half. During a lunar eclipse the Moon may turn a reddish colour. However, it is not dangerous to look at a lunar eclipse because the Moon does not make its own light.

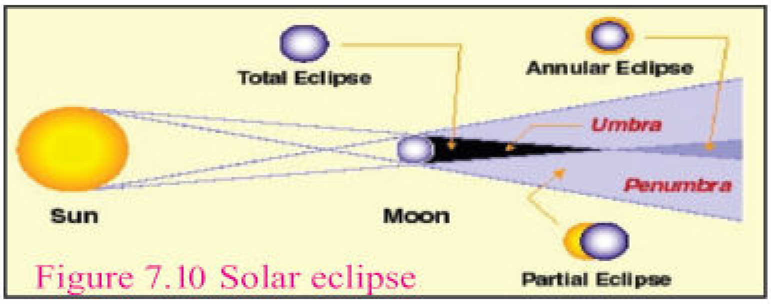


Figure 7.10 Solar eclipse

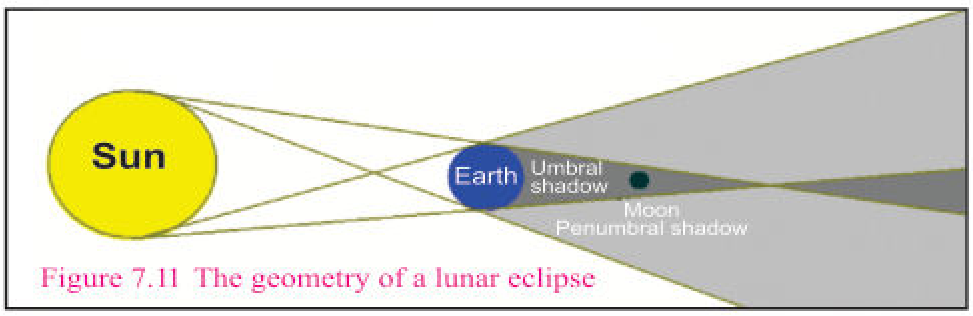


Figure 7.11 The geometry of a lunar eclipse

**Activity 4:** Demonstrating eclipses

**What I need:**

- a globe of Earth or Football (to represent the Earth)
- a small tennis ball (to represent Moon)
- a long neck glass bottle
- a projector light/Torch light source (to represent the Sun)

**What I do:**

1. Place a globe/football on the floor or on the table.
2. Keep the bottle at about 50 cm from the globe and place the ball (Moon) over its mouth.
3. Keep the projector/torch light (Sun) on the other side about 2.5 meters from the globe.
4. Now you should have an arrangement in which the bottle with the ball on its mouth comes between the globe and the projector.
5. Turn on the projector and record your observation.
6. For lunar eclipse, move the ball behind the globe.

## What I observed:

| When the ball (Moon) is between the globe/football (Earth) and the projector/torch (Sun) | When the ball (Moon) is behind the globe/football (Earth) |
|---|---|
| | |

## Activity questions:

1. How does a lunar eclipse occur?
2. How does a solar eclipse occur?
3. What can you conclude about this activity?

**Phases of the Moon**

- Identify different phases of the Moon.
- Explain the formation of a new Moon and a full Moon.

The changing shapes of the bright part of the Moon that we can see daily is called its phase.

The Moon is illuminated because it reflects the light from the Sun. The part of the Moon facing the Sun is lit up. The other part facing away from the Sun is in darkness. The Moon revolves around the Earth; we see different amounts of its surface lit during its 29.5 days.

**What causes the different phases of the Moon?**

The phases of the Moon depend on its position in relation to the Sun and the Earth. As the Moon makes its way around the Earth, we see the bright parts of the Moon's surface at different angles. These are called "phases" of the Moon.

**Teacher Note:** Demonstrate this activity. Engage students in observing, recording observations and solving activity questions.

**What are the different phases of the Moon called?**

Figure 7.12: Different phases of the Moon

The phases of the moon work in a cycle starting with the new moon.

When the Moon and the Sun are on opposite sides of the Earth,we see a full Moon. The entire side of the Moon facing the Earth is lit up by the Sun. When the Moon and the Sun are on the same side of the Earth, we see a new Moon. The side facing the Earth is in darkness.

**Do you know?**
Countries near the Equator see the crescent moon shaped like a smile?

**There are eight phases of the moon**

Figure 7.13: Eight phases of the Moon

The phases are named after how much of the moon we can see, and whether the amount visible is increasing or decreasing each day. It takes about 29.5 days to complete the cycle through all eight phases.

# Pinhole Camera

Explain the working principle of pinhole camera.

Pinhole photography is lensless photography. In a pinhole camera, a tiny hole replaces the lens. When light passes through the hole; an image is formed in the camera. Pinhole cameras are used for fun, for art, and for science.

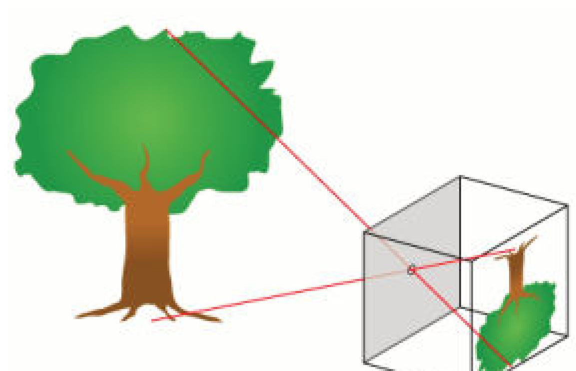


Figure 7.14: Working of pinhole camera

A pinhole camera is a small, light-tight can or box with a black interior and a tiny hole in the center of one end. It is a simple image forming device. It is a light proof box with a pinhole on one end and a piece of wax paper on the opposite inside wall. When you look through the pin hole camera, the hole lets in just few light rays from each part of the scene.

**Working principle:**

The pinhole camera works because light travels in straight lines.

Light from the top of the object passes through the pinhole and on to the screen. Light from the bottom of the object also passes through the pinhole and on to the screen to form an image. The rays keep going in straight lines and hit the butter paper screen, making an upside-down and smaller image of the scene.

Figure 7.15: Digital camera

**Do you know?** A **digital camera** is a camera that encodes digital images and videos digitally and stores them for later reproduction. Most cameras sold today are digital, and digital cameras are incorporated into many devices ranging from mobile phones to vehicles.

## Activity 5 Making a pinhole camera

### What I need:

- a shoebox ( An empty large toilet roll can also be used)
- a black chart paper/aluminium foil
- a tape
- a pair of scissors
- a square piece of tracing/butter paper

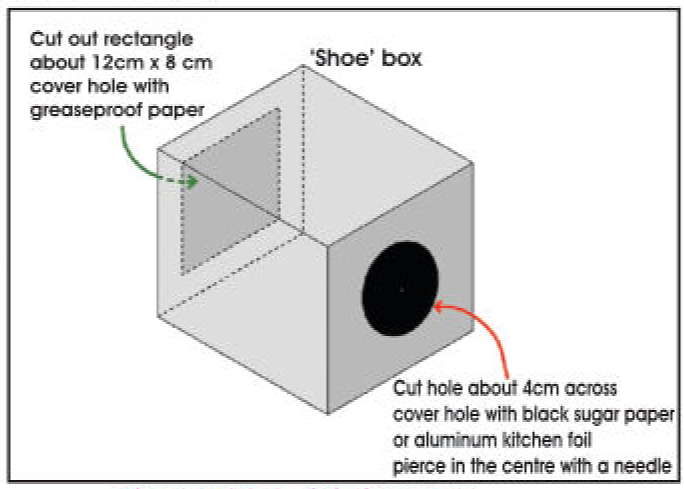



Figure 7.16: A pinhole camera

### What I do:

1. Make sure that no light can get into the box once the lid is on, so tape up any rips or joints.

2. Next, you'll need to cut two holes in opposite sides of the box so you can see right through it. Draw around a cup or mug to make the first hole. The second should be a large rectangle. (12cm x 8cm should be about the right size if you're using a shoe box.)

3. Now, cut out a square of black chart paper or tinfoil. Make sure it's bigger than the circular hole and then tape it to the inside of the box so the hole is covered.

4. Next, take your drawing pin or needle and very carefully make a small hole in the centre of the foil.

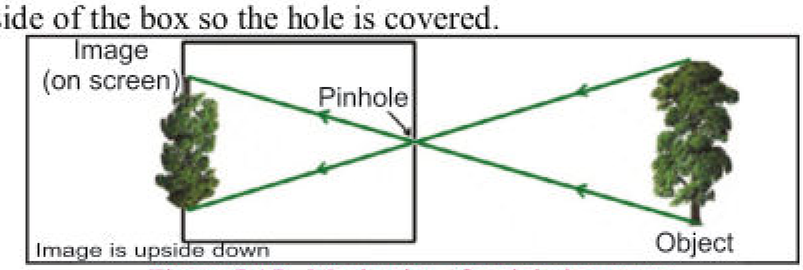



Figure 7.17: Mechanics of a pinholecamera

5. Finally, you'll need to cut out a piece of tracing paper or grease proof paper which is bigger than the rectangular hole in the other side of your box. Tape the paper to the inside of the box, making sure that it covers the hole. This will be your screen.

6. Now, tape the lid on to the box and cover it with black glaze paper, leaving the pinhole and the trace paper side.

7. Your pinhole camera is ready now.

8. Point the tinfoil/black chart paper's end of the camera towards something bright (the window is usually best) and take a look at the screen.

9. Record your observation.

Figure 7.18: A pinhole camera made up of toilet roll.

**What I observed:**

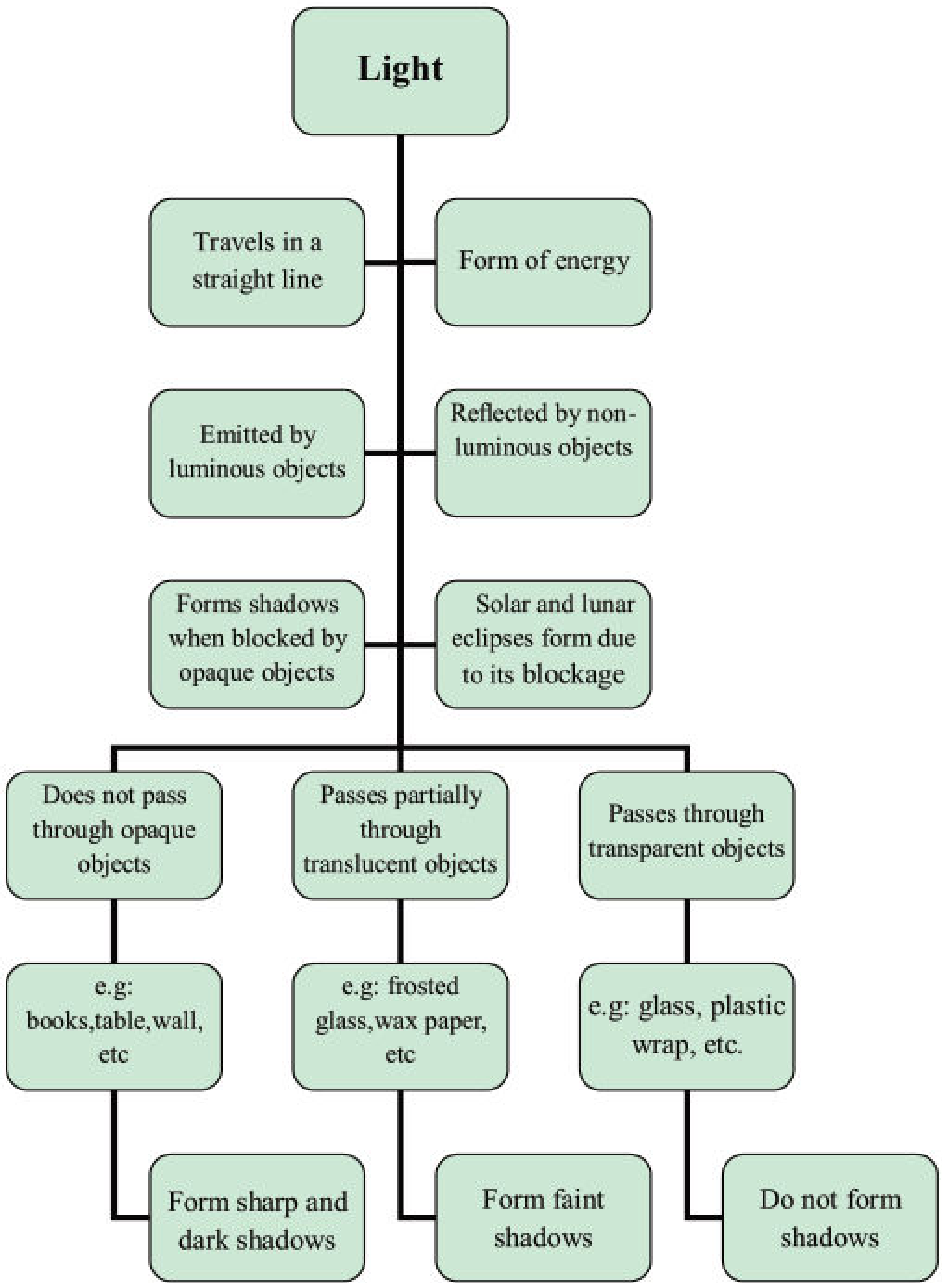
| Size of an image | Type of an image |
|---|---|
| | |

## Activity questions:

1. What is the purpose of using a tracing paper?
2. What is the purpose of wrapping the camera in a black glaze paper?
3. What might be the effect on the image if the hole is made little larger?
4. What is the working principle of a pinhole camera?

**Teacher Note:** This activity can be done in groups. Provide the necessary materials to the students and facilitate them in performing the activity.

# Summary


Light
Travels in a straight line
Form of energy
Emitted by luminous objects
Reflected by non-luminous objects
Forms shadows when blocked by opaque objects
Solar and lunar eclipses form due to its blockage
Does not pass through opaque objects
Passes partially through translucent objects
Passes through transparent objects
e.g: books,table,wall, etc
e.g: frosted glass,wax paper, etc
e.g: glass, plastic wrap, etc.
Form sharp and dark shadows
Form faint shadows
Do not form shadows

## Review questions:

**1. Fill in the blanks:**

a) _________________objects do not allow light to pass through them.

b) Butter paper is ___________________material because it scatters light.

c) Shadows are formed because light travels in a _________________line.

d) We see a _____________Moon when the Moon and the Sun are on the opposite sides of the Earth.

e) We see a _____________Moon when the Moon and the Sun are on the same side of the Earth.

**2. Tick the correct answer.**

a) Which of the following allows all light to pass through it ?

i. book
ii. newspaper
iii. glass
iv. wood

b) Which of the following is a non-luminous object?

i. Jupiter
ii. burning coal
iii. glassware
iv. fireworks

c) When the ray of light strikes on the brick wall, it is

i. passed through it.
ii. scattered by it.
iii. reflected by it.
iv. absorbed completely.

**3. Differentiate between transparent, translucent and opaque objects.**

**4. List five luminous and five non - luminous objects.**

**5. How are we able to see the Moon even though it does not give off its own light?**

**6. State differences between solar and lunar eclipses.**

**7. Draw a simple sketch to show how does non-luminous objects reflect light.**

# Project

**Properties of light**

**Kaleidoscope**

Like a microscope or telescope, the optics in a kaleido scope are used to echance our vision in some way. Vision depends on light and optics are used to reflect or bend it so that we can see in different ways. Kaleidoscopes use mirrors to reflect light into beautiful shapes and patterns.

**Let's make a kaleidoscope!**

**Materials needed:**

- Three small rectangular mirrors
- Coloured beads.
- Any adhesive tape.
- Coloured wrapping paper.
- Small empty toilet paper roll
- Pieces of transparent and transulent plastics from a used packaging.

**Procedure:**

1. Place the three mirrors together as shown. Use the long side of each mirror.

2. Build a prism by putting mirrors together, so that their reflective sides face inwards. Secure with the adhesive tape.

3. Insert this prism into a tube built out of the toilet paper tubes. This tube must be about  (1 cm) longer than the mirrors at both ends.

4. Cover one end of the prism with a round piece of transparent plastic and put coloured beads loosely on it.

5. Then cover this end of the tube with a piece of transparent plastic and seal it using the adhesive tape.

6. Close the other end of the kaleidoscope with a piece of cardboard that has a small hole (about 1/4 inch ) in its center.

7. Finally, wrap your kaleidoscope in a wrapping paper or decorate it in any other way you like.

8. Now point the toy to a source of light (make sure the light isn't too bright) and look through it. Create new unique patterns by rotating the tube. Enjoy!